

Jericho Cain, PhD

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Vancouver, WA - 98683, USA

RESEARCH FOCUS

Geometric representation learning for gravitational-wave data analysis; machine learning for astrophysical inference; matched-filter geometry; latent-space modeling; time-frequency signal processing; confusion-limited source separation for LISA; template-free detection for ground-based detectors.

ACADEMIC APPOINTMENTS

- **Portland Community College**  01/2024–Present
Portland, OR
Adjunct Physics Instructor (Weekend Teaching)
 - Teach algebra-based physics with integrated laboratory sessions to pre-medical students.
 - Incorporate discussions of advanced physics concepts and emerging roles of AI and deep learning in scientific discovery.
- **U.S. Army Research Laboratory**  08/2013– 12/2015
Adelphi, MD
Research Physicist
 - Conducted theoretical and computational research in atmospheric acoustics and turbulence.
 - Adapted ocean acoustics Green's function methods to atmospheric turbulence modeling, advancing sound-propagation prediction accuracy.
 - Contributed to the theory of statistical moments of acoustic signals in turbulent media using Python-based simulations.
- **University of Mississippi**  08/2006 – 08/2013
Oxford, MS
Graduate Researcher and Instructor
 - Participated in the LIGO project, performing data analysis, detector characterization, and Astrowatch operations during the O1 science run.
 - Developed a large-eddy simulation of atmospheric dynamics over terrain to study turbulent gust effects and ground-level pressure responses for early-warning wind turbine detection.
 - Taught calculus-based and algebra-based physics, astronomy, and associated laboratories; mentored ROTC cadets and undergraduate students.
- **Georgia State University**  08/2004 – 08/2006
Atlanta, GA
Undergraduate Lab Instructor
 - Taught algebra-based physics and physical science laboratory courses; supported curriculum delivery and student mentorship as an undergraduate instructor.

EDUCATION

- **University of Mississippi**  August 2007 – May 2013
Oxford, MS, USA
Ph.D. in Physics
 - Dissertation research focused on large-eddy meteorological modeling to simulate complete atmospheric systems and develop ground-based detection algorithms for wind gusts.
 - Conducted research with the LIGO collaboration at Hanford, WA and Livingston, LA, including interferometer operations and data analysis.
- **Georgia State University**  August 2003 – May 2007
Atlanta, GA, USA
Bachelor of Science in Physics
 - Received the Robert H. Hankla Award for Outstanding Senior Physics Student.
 - Served as an undergraduate laboratory instructor for algebra-based and physical science courses.

- [J.1] Jericho Cain (2026). **Manifold Learning for Source Separation in Confusion-Limited Gravitational-Wave Data**. *Classical and Quantum Gravity*, 2026. DOI: 10.1088/1361-6382/ae83ff
- [J.2] Jericho Cain (2026). **Template-Free Gravitational Wave Detection with CWT-LSTM Autoencoders: A Case Study of Run-Dependent Calibration Effects in LIGO Data**. *Classical and Quantum Gravity*, 2026. DOI: 10.1088/1361-6382/ae415e
- [J.3] Vladimir E. Ostashev, D. Keith Wilson, Sandra L. Collier, Jericho Cain, Sylvain Cheinet (2016). **Cross-frequency coherence and pulse propagation in a turbulent atmosphere**. *Journal of the Acoustical Society of America*, Vol. 140, Issue 1, pp. 678. DOI: 10.1121/1.4959003
- [J.4] LIGO Scientific Collaboration (incl. Jericho Cain) (2010). **Methods for Reducing False Alarms in Searches for Compact Binary Coalescences in LIGO Data**. *Classical and Quantum Gravity*, 27(16):165023. DOI: 10.1088/0264-9381/27/16/165023
- [S.1] Jericho Cain (2026). **Global Structure in Learned Latent Representations of Confusion-Limited LISA Data**. Manuscript submitted for publication in *Classical and Quantum Gravity*. arxiv:2602.20212. DOI: 10.48550/arXiv.2602.20212
- [S.2] Jericho Cain and Hayden Beadles (2026). **Mask-Based Window-Level Insider Threat Detection for Campaign Discovery**. Manuscript submitted for publication in *IEEE Transactions on Information Forensics and Security*. arXiv:2511.12845. DOI: 10.48550/arXiv.2602.11019
- [S.3] Jericho Cain (2026). **Detectability Scaling Laws for Environmental Phase Modulation in Gravitational-Wave Signals**. Manuscript submitted for publication in *PRD*. arxiv:2602.17725. DOI: 10.48550/arXiv.2602.17725
- [S.5] Jericho Cain (2026). **Gauge Freedom and Metric Dependence in Neural Representation Spaces**. Manuscript submitted for publication in *JMLR*. arxiv:2603.06774. DOI: 10.48550/arXiv.2603.06774
- [C.1] Jericho Cain (2026). **Latent-Space Density Modeling for Resolvable-Source Detection in Confusion-Limited LISA Simulations**. Presented at the *16th International LISA Symposium*, College Park, Maryland.
- [C.2] Alexander Heye, Karthik Venkatesan, Jericho Cain (2017). **Precipitation Nowcasting: Leveraging Deep Recurrent Convolutional Neural Networks**. In *Proceedings of the CUG 2017 Workshops*, pp. 1–8.
- [C.3] Jericho E. Cain, Sandra L. Collier, Vladimir E. Ostashev, David K. Wilson (2015). **Signal coherence of broadband sound propagation through a refractive and turbulent atmosphere**. *Journal of the Acoustical Society of America*, 137(4_Supplement), 2224–2224.
- [C.4] Sandra L. Collier, Jericho E. Cain, John M. Noble, David A. Ligon, W. C. Kirkpatrick Alberts, Leng K. Sim (2015). **Green's function retrieval for atmospheric acoustic propagation**. *Journal of the Acoustical Society of America*, 138(3_Supplement), 1754–1754.
- [C.5] Jericho Cain (2016). **Effects of turbulence on acoustic impulses propagating near the ground**. In *Proc. French Acoustics Congress & Conference on Vibrations, Shocks, and Noise*. Conference presentation, Apr 11, 2016.
- [C.6] Jericho E. Cain (2015). **Statistical moments of a wideband acoustic signal**. *Proceedings of Meetings on Acoustics*, Vol. 21.
- [C.7] Jericho Cain, Sandra Collier, Vladimir Ostashev, D. Keith Wilson (2014). **Spatial coherence function for a wideband acoustic signal**. *Journal of the Acoustical Society of America*, 135(4_Supplement), 2382–2382
- [C.8] Jericho E. Cain, Sandra L. Collier, John M. Noble, David A. Ligon, W. C. K. Alberts, Leng K. Sim (2016). **Inverse methods for Green's function retrieval**. *Journal of the Acoustical Society of America*, 139(4_Supplement), 1985–1985.
- [C.9] Jericho Cain, Richard Raspet (2011). **Detection of turbulence aloft by infrasonic wind noise measurements on the ground**. *Journal of the Acoustical Society of America*, 130(4_Supplement), 2437–2437.
- [C.10] Roger Waxler, Kenneth E. Gilbert, Jericho Cain (2008). **The Radiation of Microbaroms from Isolated Hurricanes over Water**. *AIP Proceedings*, 1022(1), 417–427. DOI: 10.1063/1.2838136
- [T.1] Jericho Cain (2013). **Large Eddy Simulation of Surface Pressure Fluctuations Generated by Elevated Gusts**. Ph.D. thesis, University of Mississippi. Available at: egrove.olemiss.edu/etd/771/.

RESEARCH PROGRAM: GEOMETRIC REPRESENTATION LEARNING FOR GRAVITATIONAL-WAVE DATA ANALYSIS

Learning Physics-Preserving Representations of Gravitational-Wave Data My research develops machine-learning methods that learn physically meaningful representations of gravitational-wave data while preserving the geometric structure relevant to scientific inference. Rather than replacing established techniques such as matched filtering, this work seeks to complement traditional gravitational-wave data analysis by learning low-dimensional coordinate systems that improve computational efficiency, enable new forms of source characterization, and provide interpretable representations for downstream analysis. The program spans both ground-based (LIGO) and space-based (LISA) observatories and combines differential geometry, signal processing, and machine learning into a unified research direction.

- Emphasis on representation learning guided by gravitational-wave physics rather than increasingly complex neural-network architectures.
- Development of reproducible, end-to-end analysis pipelines suitable for undergraduate and graduate research participation.
- Commitment to open science through public code, reproducible experiments, and transparent reporting of methodological limitations.

Projects:

• **Template-Free Detection for Ground-Based Detectors (LIGO)**



This research thrust develops unsupervised methods for detecting gravitational-wave signals without waveform templates by learning representations directly from detector noise. The central philosophy is to encode physically meaningful time-frequency structure prior to machine learning so that relatively simple models operate within a representation aligned with gravitational-wave morphology.

- Developed a continuous wavelet transform (CWT)–LSTM autoencoder trained exclusively on real detector noise for template-free gravitational-wave detection.
- Demonstrated competitive performance on LIGO O4 data while maintaining a fully unsupervised anomaly-detection framework.
- Identified previously unrecognized run-dependent calibration effects that substantially influenced learned representations, leading to a revised training methodology.
- Established a physics-informed framework for anomaly detection that complements matched filtering by remaining sensitive to poorly modeled or unexpected signal morphologies.
- Published in *Classical and Quantum Gravity*.

• **Representation Learning for Confusion-Limited LISA Data**



LISA presents a fundamentally different challenge than ground-based observatories because the data stream is expected to be dominated by overlapping astrophysical sources rather than instrumental noise. This research develops learned latent representations that organize the confusion foreground into physically meaningful coordinate systems for source separation and downstream inference.

- Developed manifold-learning approaches for identifying resolvable sources embedded within LISA's confusion foreground.
- Introduced latent-space geometric anomaly scores that combine reconstruction error with local manifold structure.
- Investigated whether source resolvability is governed primarily by local latent geometry or global latent structure, demonstrating that global latent density provides a more informative representation.
- Established a reproducible benchmark for evaluating geometric and probabilistic latent-space methods under identical experimental conditions.
- One manuscript accepted in *Classical and Quantum Gravity*; one manuscript currently under review.

• **Learning the Geometry of Matched Filtering**

Current work extends the geometric interpretation of matched filtering beyond Owen's local template-space metric by learning low-dimensional latent coordinate systems directly from pairwise matched-filter mismatch. Rather than approximating only infinitesimal neighborhoods, this work investigates whether the global geometry of waveform similarity can itself be learned and subsequently exploited for computationally efficient gravitational-wave data analysis.

- Learning latent coordinate systems whose Euclidean geometry preserves pairwise matched-filter mismatch relationships.
- Investigating geometric representations that may enable accelerated matched filtering, template-bank organization, visualization, clustering, and downstream inference.
- Developing a broader representation-learning framework in which physically defined similarity measures determine the learned latent geometry.
- Manuscript currently in preparation.

TEACHING EXPERIENCE

- **Portland Community College** [🌐] 01/2024 – Present
Portland, OR
Adjunct Physics Instructor
 - Physics 201: General Physics I. Topics include mechanics including statics, forces and motion energy, collisions, circular motion and rotational dynamics.
 - Physics 202: General Physics II. Topics include mechanical properties of matter, heat, waves, sound and light.
 - Physics 203: General Physics III. Topics include electricity, magnetism and radioactivity.
 - Physics 122: Stellar and Stellar Evolution. Introduces stellar astronomy, including our sun, properties of stars, and stellar evolution
 - Physics 123: Galaxies and Cosmology. Introduces star clusters, the contents of our galaxy and other galaxies, including active galaxies, and cosmology
- **University of Mississippi** [🌐] 08/2006 – 05/2011
University, MS
Graduate Student Instructor
 - Astronomy 103: Intro Astronomy of the Solar System.
 - Astronomy 104: Intro Astronomy of Stars and Galaxies.
 - Physics 211: Physics for Science and Engineering I.
 - Physics 212: Physics for Science and Engineering II.

INDUSTRY RESEARCH EXPERIENCE

- **Adobe** [🌐] 11/2022 – Present
Remote
Senior Staff Data Scientist / Research Scientist
 - Led research and development of unsupervised and semi-supervised anomaly-detection models for large-scale time-series data.
 - Designed deep learning architectures based on autoencoders and sequence models, with emphasis on interpretability and robustness to non-stationary data.
- **ConcertAI** [🌐] 10/2020 – 11/2022
Remote
Principal Data Scientist
 - Architected machine-learning pipelines for inference on incomplete, noisy, and heterogeneous clinical datasets.
 - Developed interpretable statistical and machine-learning models with explicit uncertainty and feature attribution.
- **Change Healthcare** [🌐] 08/2017 – 10/2020
Emeryville, CA
Senior Data Scientist
 - Developed machine-learning models for large-scale text and time-series analysis, outperforming manual review processes.
 - Implemented scalable training and inference pipelines using PyTorch, TensorFlow, and distributed computing frameworks.
- **Cray Inc.** [🌐] 01/2016 – 08/2017
Seattle, WA
Data Scientist II
 - Applied machine learning and high-performance computing to scientific and engineering problems, including spatiotemporal prediction.
 - Developed recurrent neural-network models for geophysical forecasting, resulting in a peer-reviewed publication.

TECHNICAL COMPETENCIES

- **Scientific Programming & Computing:** Python, C++, Fortran, MATLAB, Mathematica
- **Gravitational-Wave & Signal Analysis:** GWPy, time-frequency analysis, continuous wavelet transforms, LIGO/LISA data analysis
- **Machine Learning for Physical Sciences:** PyTorch, unsupervised learning, representation learning, anomaly detection
- **Numerical & Statistical Methods:** Bayesian inference, numerical methods, optimization, statistical modeling
- **Scientific Libraries & Visualization:** NumPy, SciPy, Pandas, Matplotlib, Jupyter
- **High-Performance & Reproducible Computing:** Parallel computing on HPC systems, version control (Git), containerized workflows
- **Research & Teaching Competencies:** Scientific writing, curriculum design, undergraduate and graduate mentorship

AWARDS AND HONORS

- **Robert H. Hankla Award** 2007 
Georgia State University
 - Presented annually to the outstanding senior physics student for academic excellence and research contributions.

ADDITIONAL INFORMATION

Languages: English (Native)
Interests and Hobbies: Astrophotography, Chess, Woodworking